

Kaden Downes

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Summary

Aspiring Electrical and Computer Engineer with hands on experience in ARM Cortex M development, FreeRTOS, PCB design, and hardware debugging. Strong background in microcontroller firmware, communication protocols, and cross disciplinary engineering projects. Always excited to learn and ready to take the next step!

Skills

- Embedded Systems
- FreeRTOS and related concepts
- Communication Protocols (UART, I2C, SPI, Wi-Fi, Bluetooth)
- Hardware Debugging (J-Link, Logic Analyzers)
- Tools: Altium, VSCode, Git, Azure DevOps
- Embedded Languages: C/C++, Rust
- Other Languages: C#, Python, VHDL, Verilog, Dart, LaTeX

Education

Bachelor of Science in Electrical Engineering	Purdue University Fort Wayne	2022–2026
Bachelor of Science in Computer Engineering	Purdue University Fort Wayne	2022–2026

Projects

Battery Powered Wet Floor Sensor (Senior Design)

Sponsor: Franklin Electric

Wrote firmware in C utilizing FreeRTOS, CMake, Ninja, and Git. Designed a PCB for the device in Altium, including a microcontroller, wireless module, sensors, and power regulation. The device communicated over Wi-Fi via TCP when certain conditions were detected, e.g., low temperature or water detected.

Experience

Advanced Repair Agent	Geek Squad — Fort Wayne	Nov 2021–Present
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- Performed detailed fault analysis to determine software vs. hardware causes and carried out component level repairs on multiple device platforms.
- Developed automation scripts using PowerShell and Bash to streamline workflows.
- Closed 137 tickets every month, averaging 1.2 tickets per hour with 140% labor efficiency.
- Communicated with clients and other agents as to the status of and next steps for work orders.
- Responsibly handled sensitive customer data, including storing and transferring personal files.
- Achieved 100% customer satisfaction according to Apple repair surveys.